

GENOMIC ENGINEERING, BIOINFORMATICS & SYNTHETIC SYSTEMS

New Egyptian Baccalaureate (EG-Bac) Advanced Biotechnology Syllabus

ACADEMIC YEAR: 2024/2025 | GRADE 12 (EG-BAC ADVANCED TRACK)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Advanced Egyptian Baccalaureate textbook in Genomic Engineering, Computational Bioinformatics & Synthetic Biology: Illumina reversible terminator chemistry, Oxford Nanopore ionic current translocation, RNA-Seq TPM transcriptomic normalization, Needleman-Wunsch global dynamic programming, Smith-Waterman local alignment, Karlin-Altschul BLAST E-value statistics, AlphaFold 3D protein structure prediction with Evoformer modules, CRISPR-Cas9 RuvC/HNH endonuclease cleavage, Cytosine/Adenine Base Editing (CBE/ABE), pegRNA prime editing, synthetic biology toggle switches, volumetric oxygen transfer kLa modeling, CAR-T scFv cell therapy engineering, and Cartagena Protocol biosafety governance.

CORE CURRICULUM HIGHLIGHTS:

- Flow cell bridge amplification gigabase throughput and RNA-Seq TPM normalization
- Dynamic programming recurrence matrix equations and AlphaFold pLDDT confidence scores
- Cas9 PAM recognition kinetics, base editing deamination, and prime editing pegRNA design
- Bioreactor volumetric oxygen transfer rate (OTR) hydrodynamics and CAR-T cytolytic activity

APPROVED CHAPTER MODULES:

- Next-Generation Sequencing, Nanopores & Transcriptomics	pp. 1 82 (3 core units)
- Computational Bioinformatics, Sequence Alignment & AlphaFold	pp. 83 164 (3 core units)
- CRISPR Genome Editing, Base/Prime Editing & Synthetic Circuits	pp. 165 246 (3 core units)
- Bioprocess Fluid Dynamics, CAR-T Engineering & Global Biosafety	pp. 247 328 (3 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Advanced Genomics, CRISPR & Bioinformatics Mathematical Compendium (8p - und
Full Ministerial Textbook: 328 Pages - 30.6 MB (Direct on MOE Portal)
Publisher: Egyptian Ministry of Education & Technical Education
Official Portal: <https://moe.gov.eg/elearning/egbac/advanced-biotechnology-g12> - Code:
Legal Deposit: 2025/14892 - ISBN: 978-977-12-2376-71



MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

Advanced Egyptian Baccalaureate textbook in Genomic Engineering, Computational Bioinformatics & Synthetic Biology: Illumina reversible terminator chemistry, Oxford Nanopore ionic current translocation, RNA-Seq TPM transcriptomic normalization, Needleman-Wunsch global dynamic programming, Smith-Waterman local alignment, Karlin-Altschul BLAST E-value statistics, AlphaFold 3D protein structure prediction with Evoformer modules, CRISPR-Cas9 RuvC/HNH endonuclease cleavage, Cytosine/Adenine Base Editing (CBE/ABE), pegRNA prime editing, synthetic biology toggle switches, volumetric oxygen transfer kLa modeling, CAR-T scFv cell therapy engineering, and Cartagena Protocol biosafety governance. Flow cell bridge amplification gigabase throughput and RNA-Seq TPM normalization Dynamic programming recurrence matrix equations and AlphaFold pLDDT confidence scores Cas9 PAM recognition kinetics, base editing deamination, and prime editing pegRNA design Bioreactor volumetric oxygen transfer rate (OTR) hydrodynamics and CAR-T cytolytic activity

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	45% Understanding & Application Calculations, proofs & models	25% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
---	---	--

3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal: <https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Next-Generation Sequencing, Nanopores & Transcriptomics	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
Computational Bioinformatics, Sequence Alignment & AlphaFold	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
CRISPR Genome Editing, Base/Prime Editing & Synthetic Circuits	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
Bioprocess Fluid Dynamics, CAR-T Engineering & Global Biosafety	7 wks	15 pts	25.0%	3 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: NEXT-GENERATION SEQUENCING, NANOPORES & TRANSCRIPTOMICS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Next-Generation Sequencing, Nanopores & Transcriptomics.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Next-Generation Sequencing, Nanopores & Transcriptomics.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: COMPUTATIONAL BIOINFORMATICS, SEQUENCE ALIGNMENT & PROTEIN FOLDING

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Computational Bioinformatics, Sequence Alignment & AlphaFold.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Computational Bioinformatics, Sequence Alignment & AlphaFold.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: CRISPR GENOME EDITING, BASE/PRIME EDITING & SYNTHETIC

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of CRISPR Genome Editing, Base/Prime Editing & Synthetic Circuits.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of CRISPR Genome Editing, Base/Prime Editing & Synthetic Circuits.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

GENOMIC ENGINEERING, BIOINFORMATICS & SYNTHETIC SYS

Fundamental Identity: $\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$

Equilibrium Balance Criterion: $\sum F_{\text{net}} = 0$, $\sum M_{\text{pivot}} = 0$

Continuity & Conservation Standard: $\text{div}(\mathbf{J}) + \text{partial } \rho / \text{partial } t = 0$

Normalized Standard Score: $Z = (X - \mu) / \sigma$

PROCEDURAL ASSESSMENT & TRANSFORMATION RULES

Linear Operator Transformation: $T(\alpha * u + \beta * v) = \alpha T(u) + \beta T(v)$

Harmonic Synthesis Mean: $H = n / (\sum_{i=1}^n 1/x_i)$

Sample Variance Boundary: $s^2 = (1 / (n - 1)) * \sum (x_i - \bar{x})^2$

Curricular Efficiency Ratio: $\eta = (\text{Output_actual} / \text{Input_stand})$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

Official Document Code: EGBAC-BIOTECH-G12
Curriculum Cycle: 2025/2026 Academic Year | National Edition
Official Portal: <https://moe.gov.eg/ar/elearning-content/>
All rights reserved (c) 2025 Ministry of Education & Technical Education - Egypt.
Unauthorized commercial reprinting or distribution without ministerial sanction is prohibited.